

PCO_{NA} compared to as-transferred samples imply that strain effects are negligible^{14,33}. X-ray photoelectron spectroscopy (XPS) shows only peaks associated with Ce/Pr, with no evidence of other species such as Sr and Cr which would cause degradation as shown in **Supplementary Fig. 1**. Finally, Au current collectors were sequentially deposited on both ends of the PCO_{NA} for evaluating in-plane conductivity. The distance between the two current collectors and the width of the model platform were fixed to 1 mm and 1 cm, respectively.

Surface-infiltrated oxide acidity effects on PCO_{NA} electronic properties

To investigate the effects of binary oxide acidity on the electrical conductivity of PCO_{NA}, two PCO_{NA} specimens with different binary oxides infiltrated onto their surfaces were prepared and deposited on single crystal sapphire (Al₂O₃) substrates. The in-plane conductivity of uninfiltrated PCO_{NA} as well as PCO_{NA} infiltrated by SiO₂ (acidic) and by Li₂O (basic) were measured utilizing AC impedance spectroscopy, respectively. Additionally, to explore the potential for recovering degraded conductivity, PCO_{NA} with serial infiltration of Si-species followed by Li-species was also prepared and evaluated²⁴.

The primary objective of this study was to monitor conductivity changes induced by heterointerfaces through the surface infiltration of binary oxides with different acidity, rather than the initial focus on the oxygen exchange rate as reported previously^{21,34,35}. Hence, we attempted to achieve full PCO_{NA} coverage by using a concentrated solution (0.2 M) to maximize coverage and as shown in SEM images in **Supplementary Fig. 2**, the coverage is complete. It is important to note that the electrical conductivities of the insulating binary oxide infiltrants used in this study are negligible compared to that of the PCO nanowires (**Supplementary Table 2**). The measured impedance spectra are therefore assumed to solely represent PCO_{NA} electrical pathways. AC impedance spectra were obtained between 450 – 650 °C to extract the in-plane conductivity of both the uninfiltrated and infiltrated samples (**Supplementary Fig. 3**).

As illustrated in **Fig. 2a** and **2b**, Li-infiltration resulted in nearly three orders of magnitude decrease of resistance (see inset), while Si-infiltration caused only a slight increase in resistance, both relative to the uninfiltrated PCO_{NA}. Arrhenius plots of in-plane conductivity under each infiltration condition, along with the corresponding activation energy (E_a) values, are summarized in **Fig. 2c**. Consistent with the impedance plots, Li-infiltrated PCO_{NA} exhibited approximately three orders of magnitude higher in-plane conductivity compared to uninfiltrated PCO_{NA} over the full temperature range, with a same nominal E_a of 0.85 ± 0.023 eV.

However, only 1.5 times lower conductivity was measured in Si-infiltrated PCO_{NA} relative to the uninfiltrated reference with similar E_a (0.89 ± 0.025 eV). A similar trend was also observed in serial infiltration with Li-species following Si infiltration, aimed at recovering conductivity, as shown in **Fig. 2b** (see also inset). Subsequent Li infiltration on Si-infiltrated PCO_{NA} not only restored, but exceeded the initial conductivity by 100-fold with a reduced E_a of 0.63 ± 0.065 eV. Although serial Li infiltration did not achieve as much improvement in conductance as sole Li infiltration, given the pre-coating effects of SiO₂, it once again verified the strong positive impact of basicity on already degraded electrodes.

Insulating substrate acidity effects on PCO_{NA} electronic properties

We next explored the impact of the choice of insulating substrate on the electronic properties of our PCO_{NA}. This is not only of relevance to further validate the observed trends of heterointerface acidity on the electrical conductivity of PCO_{NA}, but also considering that diverse insulating substrates, such as Al₂O₃ or MgO, are commonly used to study the electrical properties of deposited oxide thin films. Generally, one employs such substrates to support controlled film growth while minimizing the potential for short-circuiting of the supported films through the substrate. The potential impact of their interfacial properties on the overlaying films' conductivity, however, is generally ignored based on the assumption that transport along such interfaces would have an insignificant impact compared to transport through the bulk (R_1). It is noteworthy that most of these insulating substrates each possess their own distinct relative Smith acidities, and thereby work functions, which could presumably result in inducing differing space charge phenomena at these interfaces³⁶. PCO_{NA} were fabricated on both Al₂O₃ and MgO substrates (**Supplementary Fig. 4**), expected to exhibit acidic and basic characteristics respectively relative to PCO₂₀. This enabled us to assess any alterations in the in-plane conductivity induced at the heterointerface between the substrates and PCO_{NA}.

The AC impedance spectra of PCO_{NA} fabricated on the two different substrates were obtained for temperatures between 450 – 650 °C as shown in **Supplementary Fig. 3 and 5**. By comparison of these two impedance spectra at the same measurement temperature of 650 °C (**Fig. 3a**), it becomes immediately obvious that PCO_{NA} fabricated on Al₂O₃ is over an order of magnitude more resistive than the same structure fabricated on MgO. Arrhenius plots of in-plane conductivity are shown in **Fig. 3b** along with a listing of their respective E_a values. Consistent with the impedance plots, PCO_{NA} on the MgO substrate exhibited approximately 10

times higher in-plane conductivity compared to PCO_{NA} on the Al_2O_3 substrate over the full temperature range, with a nominally lower E_a of 0.78 ± 0.037 eV vs. 0.85 ± 0.007 eV. Since all other conditions were identical except for the choice of substrate, it is reasonable to assume that the differences in conductivity between these two samples can be attributed to the relative acidity of the substrates. With this insight, it can be argued that the selection of a suitable substrate must be considered before depositing nanostructured semiconductors or MIECs to avoid such heterointerface acidity effects. Indeed, we suspect that many reports of thin film oxide conductivity have been impacted by the relative substrate acidity of the supporting substrate and thus need to be reevaluated or appropriately normalized.